

Foundation (Jalapeno)

Q1. Solve for x in $2^x = 8$

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$
- (E) $x = 6$

Q2. If $3^x = 27$, what is x ?

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q3. A company's stock price is modeled by $P(t) = 2^t$, where t is time in years. If $P(t) = 16$, what is t ?

- (A) $t = 2$
- (B) $t = 3$
- (C) $t = 4$
- (D) $t = 5$
- (E) $t = 6$

Q4. Solve for y in $4^y = 64$

- (A) $y = 1$
- (B) $y = 2$
- (C) $y = 3$
- (D) $y = 4$
- (E) $y = 5$

Q5. A savings account earns interest according to $A(t) = 5 \cdot 2^t$, where t is time in years. If $A(t) = 20$, what is t ?

- (A) $t = 1$
- (B) $t = 2$
- (C) $t = 3$
- (D) $t = 4$
- (E) $t = 5$

Q6. If $x^2 = 2^4$, what is x ?

- (A) $x = 2$
- (B) $x = 4$
- (C) $x = 6$
- (D) $x = 8$
- (E) $x = 10$

Q7. Solve for z in $2^z = 32$

- (A) $z = 3$
- (B) $z = 4$
- (C) $z = 5$
- (D) $z = 6$
- (E) $z = 7$

Q8. A tax on a stock is calculated as $T(x) = 3 \cdot 2^x$, where x is the number of shares. If $T(x) = 24$, what is x ?

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Open Ended Questions (FRQ)

Q9. Solve for x in the equation $x^3 = 2^6$. Show all work and explain your reasoning.

Q10. A company's revenue is modeled by the equation $R(t) = 2^t \cdot 1000$, where t is time in years. Find the revenue after 4 years and explain the calculation.

Chef's Tips

- Tip 1: Encourage students to check their work by plugging the answer back into the equation.
- Tip 2: Remind students to use the properties of exponents when solving equations.
- Tip 3: Emphasize the importance of labeling units and explaining reasoning in free-response questions.